Overnight Research: Hypotheses, Protocol and Results

fabv2 research notes · 6 November 2026 · all figures are indicative; the cited report is authoritative

Summary

Only one overnight effect is solid on our data: the **overnight premium**. On NQ over 2016–2026, close-to-open returns +4.8 bp/day (t = 3.1), almost all of it in the Asian session. The rule that captures it (12-1 momentum, entry 20:00 ET, stop 2 / target 5 ATR) is validated out of sample on 2011–2015 (+0.116 R/trade, p = 0.0015).

- **What holds**: being long NQ overnight, with alpha of about +6%/year and a beta of 0.09.
- **What does not**: almost every filter and refinement (stops, entry time, VIX, macro announcements, selling pressure, 3-night filter). None beats a random filter of the same size or survives out of sample.
- **What remains open**: the rebound after a sharp 3-night decline, and the concentration of gains on NFP/CPI nights. Both are deferred to the holdout frozen in pre-registration no. 26 (first read on 2028-02-29).

Global trial counter at the time of writing: 1,742 (about 610 effective trials, see Statistical inference).

The hypothesis: a premium for carrying overnight risk

Thesis. Overnight returns compensate those who agree to carry the inventory that others do not want to hold after the close. They provide liquidity for end-of-day rebalancing, then remain exposed during hours when unloading is expensive or impossible. The premium is the price of that service.

Motivation

US indices earn most of their return outside regular trading hours, while most volume and information arrive during the session. An ordinary risk premium should be spread according to the risk borne. Yet, in the literature, the US session carries the majority of daily variance, and on our data it earns less than the night. A mechanism specific to the night is therefore needed.

Proposed mechanism

1. **Constrained closing flows.** At 16:00 ET, some participants must trade at the close whatever the price: index funds and ETFs rebalancing, leveraged ETFs resetting their exposure, market-on-close orders. Arbitrageurs also cut positions before the night, because margins and borrowing fees are higher overnight (Bogousslavsky). On average this flow is net selling.
2. **Absorption by capital-constrained intermediaries.** Market makers and overnight risk bearers take the other side. They end up holding long inventory they did not choose.
3. **A night with no cheap exit.** Globex stays open for futures, but liquidity is thin: the share of volatility exceeds the share of volume, and each contract moves the price more. Unloading before Europe, and then the US open, arrive would be costly. The holder therefore stays exposed to overnight news.
4. **Required return.** To accept this inventory and risk, the intermediary buys at the close at a discount. The price recovers when liquidity returns (Europe, then 9:30): this is the overnight premium.

This is the inventory-risk mechanism of Boyarchenko, Larsen and Whelan (*The Overnight Drift*, RFS 2023). Adrian et al. (*Intraday Market Making with Overnight Inventory Costs*) provide the model: an overnight inventory cost forces liquidation at the end of the session and creates a premium for whoever carries the risk.

Testable predictions, and what our data say

Prediction of the mechanism	Our data	Status
P1. Returns accrue overnight, not during the session	NQ close-to-open +4.81 bp/day (t 3.1); US session +3.02 (t 1.4), unstable	Supported
P2. The premium grows with the risk borne (higher beta, Hendershott et al.)	NQ +4.81 vs ES +3.45 and YM +3.28; the mom rule works on NQ only	Consistent
P3. Asymmetry: strong rebound after heavy selling, little reversal after rallies	Lowest decile of r3: next night positive in 6 samples out of 6; after rallies the night is flat, not negative	Watch list (uncorrected)
P4. The premium tracks the closing sell imbalance	Proxied by close < VWAP 15:15–16:15 (no. 16): NQ +0.051 R (p 0.11), ES p 0.30, YM reversed	Not shown with this proxy
P5. The premium is larger when overnight risk is larger	NFP/CPI nights: +0.30 R vs +0.05	Watch list
P6. Unloading happens when liquidity returns	On NQ most of it comes from 16:00 → 3:00 ET (+4.35 bp, t 3.7), which ends at the European open; Europe itself +0.49	Partly supported

What is missing to conclude

A direct test of P4 requires the signed volume at the close (tick-by-tick trades, about $1,275 for NQ). Without it, inventory risk cannot be separated from other explanations consistent with P1 and P2: a beta premium concentrated overnight (Hendershott et al.), overnight vs. daytime clienteles (Lou, Polk and Skouras, *A tug of war*), or attention-driven buying at the open (Berkman et al.). Finally, the same authors describe in 2026 a weakening drift: the mom rule is flat on the early holdout read, though two months are not enough to tell.

Setting

We study US index futures (NQ, ES, YM) and gold (GC, SI) on one-minute bars, with vectorbt. Times are New York (ET).

Item	Choice	Detail
Data	Databento 1 min	2016-01 → 2026-08-28; 2010–2015 bought later as a temporal out-of-sample; London Strategic Edge after 2026-08-28
Rolls	Ratio adjustment	Rule from run_sessions.py (the cached *_adj.parquet misses about 10 rolls per asset)
Sessions	Asia, Europe, US	Asia = 16:00 NY → 08:00 London; Europe = 08:00 London → 9:30 NY; US = 9:30 → 16:00. Asia + Europe = close-to-open
Night for mom	20:00 → 9:30 ET	Early exit at the stop (2 ATR M15) or target (5 ATR M15)
Night for 3n / rebound	20:05 → 03:00 ET	Open to open; r3 = sum of the 3 previous nights
Standard cost	1 tick + $5 round trip	Variants: real bbo-1m spreads, CFD (2 pts NQ), MNQ (1 tick + $1.50)
Unit	R	1 R = stop distance, i.e. 2 ATR M15 at entry

An adjusted futures return is an excess return (index + dividends − rates), not the price return of the cash index.

Protocol

Every confirmatory test is written and committed (PREREGISTRATION_<n>.md) before any computation. Everything else is labelled exploratory and cannot validate a rule.

1. **Pre-registration**: rule, data, cost, statistical test and threshold fixed in advance.
2. **Trial counter**: every rule tested increments a global counter. It is the N of the Deflated Sharpe Ratio (criterion DSR > 0.95) and tells how many false positives to expect.
3. **Out of sample**: IS 2016–2022, OOS 2023 → 2026-08, then 2010–2015 as an unseen temporal OOS. Another asset over the same period (ES, YM) is a weak OOS, since the indices are correlated.
4. **Benchmarks**: every overnight rule is compared to a plain long over the same hours, and every filter to 500 random filters of the same size.

5 **Multiple-testing correction**: Holm or BH within each family; Reality Check, PBO and bootstrap as complements.

6 **Single holdout**: data after 2026-08-28, plan frozen in no. 26. The window 2026-08-29 → 11-03 was read early (amendment A), so the official holdout starts on 2026-11-04.

Holdout reading rule: reads on 2028-02-29 and 2029-08-31 (success only if $z \geq 3.0$, Haybittle–Peto threshold), verdict on 2031-08-29 with Holm. Until then, no signal or return computation on these data.

Statistical inference

How each t-statistic and p-value is computed, and whether serial dependence, heteroskedasticity and the number of trials change the conclusions (exploratory check, reports/inference_overnight_report.md; no new rule, trial counter unchanged).

Methods

- **Pre-registered tests of mom (no. 10–12)**: confidence intervals and one-sided p-values by circular moving-block bootstrap (blocks of 10 trades or days, 2,000 draws). Alpha (no. 12): block bootstrap on α + residuals.
- **3n, M2 and the holdout (no. 24–26)**: HAC Newey–West t-statistic with 5 lags.
- **Session means** (e.g. NQ +4.81 bp/day): originally a plain iid t; re-estimated below with Newey–West (Bartlett kernel, 5 lags and the automatic rule $\text{floor}(4(T/100)^{2/9}) = 8$ lags) and a block bootstrap.
- **Dependence diagnostics**: Ljung–Box (5 and 10 lags) on returns and squared returns, ARCH-LM (5 lags), first-order autocorrelation, excess kurtosis.
- **Multiple testing**: Holm or BH within each family, White's Reality Check, and the Deflated Sharpe Ratio with N = global trial counter; here also with an effective N.

iid vs HAC t-statistics

Series (2016–2026 unless stated)	Mean	t iid	t NW(8)	p NW	p block	ρ1
NQ close-to-open	+4.82 bp/day	3.12	3.51	0.0004	0.0005	−0.03
NQ Asia	+4.35	3.66	4.27	0.00002	< 0.0005	−0.13
NQ US session	+3.02	1.42	1.60	0.11	0.11	−0.10
ES close-to-open	+3.45	2.80	2.99	0.003	0.002	−0.03
YM close-to-open	+3.28	2.74	2.88	0.004	0.0045	−0.02
mom NQ (return per night)	+3.18 bp	2.92	3.16	0.0016	0.0035	−0.02
mom NQ OOS 2011–2015 (R)	+0.116 R	2.79	2.87	0.004	0.0045	−0.04

Two-sided p-values. The HAC t is **larger** than the iid t everywhere: first-order autocorrelation is negative (about −0.13 in Asia, −0.10 in the US session), which lowers the long-run variance of the mean. The iid t was conservative. Newey–West and the block bootstrap agree; no conclusion changes.

Serial dependence and heteroskedasticity

- **Strong volatility clustering** in every return series in basis points: ARCH-LM(5) between 144 and 1,035, $p < 1e\text{-}28$; excess kurtosis 5 to 15. This is why HAC or block-bootstrap inference is required.
- **Weak but real serial correlation** in the Asian and US sessions (Ljung–Box(5) $p < 1e\text{-}5$); close to zero for the full close-to-open return.
- **Measured in R (ATR-based stop), mom shows no ARCH effect** ($p = 0.47$) and no fat tails: ATR sizing already normalises risk, consistent with the rejection of volatility-based sizing in no. 15.

Effective number of trials

Daily net return series are stored for 560 of the 1,742 trials. On their correlation matrix, the Li–Ji eigenvalue method gives $\mathbf{M_{eff}}$ **= 196** (ratio 0.35); Nyholt gives 535 (known to overstate with weak average correlation); 172 eigenvalues explain 95% of the variance. Most pairs are nearly independent (median |ρ| 0.027) but a few clusters are highly redundant (largest eigenvalue 76: the overnight premium tested at every hour and stop variant). Extrapolating the ratio gives **≈ 610 effective trials** out of 1,742.

Deflated Sharpe Ratio with raw and effective N

Series	Annual Sharpe	N	SR0 (observed V)	DSR	SR0 (V = 1/T)*	DSR*
mom NQ 2016–2026	0.88	1,742	1.78	0.001	1.03	0.30
mom NQ 2016–2026	0.88	≈ 610	1.63	0.005	0.94	0.42
NQ close-to-open	0.96	1,742	1.78	0.004	1.03	0.41
NQ close-to-open	0.96	≈ 610	1.63	0.016	0.94	0.53

* Control added after the first computation: V set to the pure sampling variance of a Sharpe ratio.

Moving from 1,742 to about 610 trials does not change the verdict: N enters the threshold through a log-type term, so dividing it by three lowers SR0 by only about 9%. What drives SR0 is V, the dispersion of Sharpe ratios across trials. Neither mom nor the raw overnight premium passes the DSR against the global counter, raw or effective. The evidence for mom rests instead on the pre-registered out-of-sample test of no. 11 (2011–2015, a single hypothesis, one-sided p = 0.0015), where the relevant N is 1, and on the holdout of no. 26.

Not done: Romano–Wolf stepM or Hansen's SPA over a full family, and an effective N over all 1,742 trials, both of which require the return series of the 927 batch-1 tests, which were not stored.

Hypotheses tested

About twenty overnight hypotheses have been tested since October 2026. They come from the literature (SSRN batch 1), two strategies passed on to us, and our own reading of the data.

Hypothesis	Origin	Test	Verdict
Index returns accrue outside the session (overnight premium)	Overnight literature	Sessions 2016–2026, ES/NQ/YM	Confirmed
Long/short NQ 20:00 → 9:30 by 12-1 momentum, SL 2 / TP 5 ATR (“mom”)	Batch 1, chosen after the fact	no. 10, 11, 12	Yes on NQ, no on ES and YM
mom is more than beta	—	no. 12	Yes, α +6.2%/yr
Optimised stops and entry time improve mom	—	no. 12, entry times	No (improve the tail, not the Sharpe)
“Stay flat” filters (Jones, previous night, VOL, MA20) reduce drawdown	Jones, Abdi, Berkman, Vojtko	no. 13	No
Avoid NFP/CPI nights	Risk intuition	no. 14	No, reversed
VIX filters; size by volatility	Donninger, Vojtko	no. 15	No
Close below VWAP 15:15–16:15 (selling pressure)	BLW	no. 16	No
VWAP execution	—	no. 17	Not retained
Stop at the evening range 17:00 → 20:00	Us	Exploratory	No
Sign of one session predicts the next	Us	Exploratory	No
Long NQ 20:05 → 03:00 if the previous 3 nights ≤ 0 (“3n”)	Strategy passed on	no. 24	No, out of sample
The higher r3, the more the night falls	Us	Exploratory	No
Rebound after a sharp decline (r3 ≤ −50 bps)	Us, reading deciles	Exploratory	Watch list
Gold: evening reopening 17:00 → 9:00 Shanghai	Gold sessions	Exploratory	Yes gross, weak net
Gold: noisy price before Europe (fade early Globex)	Fattinger–Ziegler	Exploratory	No

Hypothesis	Origin	Test	Verdict
Gold: the 17:00 → 5:00 move continues through the day	Fattinger–Ziegler	Exploratory	Watch list

Reports: reports/prereg<n>_report.md, and reports/ for exploratory work.

What is significant

1. The overnight premium (confirmed)

Over 2016–2026, close-to-open is positive and significant on all three indices: NQ +4.81 bp/day (t 3.12), ES +3.45 (t 2.80), YM +3.28 (t 2.74). The US session is positive but not significant (t 1.3 to 1.4) and very unstable from year to year. On NQ, most of it comes from Asia (+4.35 bp, t 3.66); Europe is close to zero (+0.49).

2. The NQ overnight rule "mom" (candidate, validated out of sample)

Measure	Value	Source
Out of sample 2011–2015 (no. 11)	+0.116 R/trade, p 0.0015	reports/prereg11_report.md
Alpha, beta (no. 12)	α +6.2%/yr [+2.5; +9.7], β 0.09	reports/prereg12_report.md
With real bbo-1m spreads	+0.076 R/trade [+0.033; +0.116]	reports/couts_reels_nq_nuit_report.md
2011-07 → 2026-11-03	3,831 nights, +0.087 R/night, Sharpe 1.01, t 4.0, max DD −25.4 R	results/mom_perf/

Important caveat: over 2011–2015, 100% of trades were long. What is validated is therefore "long NQ overnight", not the momentum filter. The filter cuts the 2022 loss (DD −10.7% vs −25.8% always long), but on only 422 nights: consistent, not proven.

3. Gold: the rise happens at the evening reopening (confirmed gross)

On GC, the 17:00 → 9:00 Shanghai session earns +2.0 bp gross/day (t 3.5, Holm 0.003), positive in all 3 sub-periods. Net of costs, +1.1 bp remains (t 2.0): real, but barely tradable. London and COMEX are flat.

4. Watch list (significant gross, not after correction)

- **Rebound after a sharp decline**: if r3 ≤ about −50 bps (lowest decile), the next night is positive in 6 samples out of 6 (NQ +17.3 bps t 3.7; ES +15.9 t 3.3; YM +11.1 t 2.5 over 2016–2026, and also positive over 2010–2015). Found by reading deciles, hence outside the Holm family: this is H4 of the holdout.
- **NFP/CPI nights**: +0.30 R vs +0.05 for other mom nights; about 40% of R in 9% of nights. Found by reversing the hypothesis of no. 14; does not generalise to other announcements.
- **Gold 16:00 → 19:00 NY**: +1.4 bp net (t 2.7), chosen after the fact, flat in 2016–2019.
- **Gold, continuation**: the 17:00 → 5:00 move extends through the rest of the day (slope 1.13, Holm 0.02), but only over 2023–2026 and found among 23 cut-offs.

Sources: wiki/phenomenes.md, results/sessions/RAPPORT.md, reports/gc_sessions_report.md, reports/nq_nuit_r3_pente_report.md.

What failed, and why

The common lesson: overnight, almost everything that "works" is the overnight premium counted again under another name. A filter adds something only if it beats the nights it skips, and none has done so robustly.

Idea	Key figure	Why it failed	Reopen only if
The 56 net-positive ICs of batch 1	51 of 56 = same premium	About 23 false positives expected out of 927 tests	—
mom on ES and YM (no. 10)	ES Holm p 0.25	Effect 3–5× weaker than on NQ; YM negative 2021–26	—
Stops and entry time (no. 12)	+0.14 Sharpe [−0.22; +0.50]	Gain in the tail, not the Sharpe; 20:00 wins through exposure time	Tail-risk objective
"Stay flat" filters (no. 13)	Below 95th pct of 500 random filters	None cuts DD by 30% while keeping 80% of R	Written rationale + holdout test

Idea	Key figure	Why it failed	Reopen only if
Avoid NFP/CPI (no. 14)	+0.30 R vs +0.05	These are the best nights	Never in this direction
VIX filters, vol sizing (no. 15)	DD −1% only	The ATR stop already normalises risk	New non-price, non-VIX data
VWAP selling pressure (no. 16)	NQ +0.05 R, p 0.11	ES not significant, YM reversed	True tick-level signed volume
Stop at the evening range	−0.10 to −0.03 R net	Median stop 0.86 ATR: 60% stopped, cost 0.13 R	Near-zero cost
Session sign → next session	Europe → US r −0.003	No exploitable correlation	Flow data with a mechanism
3n strategy (no. 24)	+0.016 R, p 0.32 on 2010–15	Skipped nights also earn +0.092 R: it is the premium, not the filter	Holdout, frozen rule
High r3 → falling night	$\|t\| \le 1.5$, Holm 1.0	After rallies the night is flat, not negative	Never as a short
Gold: GC close-to-open with ATR SL/TP	−0.9 bp/trade net	Negative in every variant	—
Gold: fade early Globex	Slope 0.68 at 19:00, Holm 0.97	No efficiency drop before Europe	Evening bid/ask data

Details: wiki/Failed Ideas/ledger.md.

Limitations and next steps

The main limitation: mom was chosen after the fact among hundreds of rules, and its out-of-sample validation covers long nights only.

- **Only one real bear market** (2022) in the sample: the momentum filter's behaviour in a prolonged decline is not demonstrated.
- **Concentration**: about 40% of R come from 9% of nights (NFP/CPI). The result depends on few events.
- **Independence**: mom and 3n are correlated (+0.35 per night) and share the same mean reversion. Any combination must be pre-registered.
- **Early holdout read** (2026-08-31 → 11-03, exploratory): mom −0.023 R/night over 44 nights, 3n +0.379 R over 23 trades (p 0.087), rebound spread +19.8 bps (p 0.13). No conclusion is possible at this size.

Next steps:

- Download NQ monthly without computing anything (official holdout since 2026-11-04).
- First holdout read on 2028-02-29 (H1 mom, H3 3n, H4 rebound; success only if $z \ge 3.0$).
- Pre-register on the holdout the gold window 16:00 → 19:00 and the 17:00 → 5:00 continuation.
- Pre-register over-weighting NFP/CPI nights, with a written mechanism.

In parallel, mom runs on the FTMO demo account A (plan adopted on 2026-11-04); this is an execution test, not statistical evidence.